

Mid - Term Examination, Even Semester 2021-22

B. Tech (CSE), 3rd Year, 5th Semester

BCSC0011: Theory of Automata & Formal Languages

Time: 2 Hours

Maximum Marks: 30

Section-A

Attempt All Three Questions.

[3 x 2 = 6]

1. Draw a deterministic finite automaton (DFA) that recognizes the language over the alphabet $\{0, 1\}$ consisting of all those strings that contain an odd number of 1's.
2. Find all strings in $L((a + b)^*b(a + ab)^*)$ of length less than four.
3. Draw a NFA for the language $L = \{ w \in \Sigma^* \mid w \text{ contains the substring } 0101, \text{ i.e., } w = x0101y \text{ for some } x, y \in \Sigma^* \}$

Section- B

Attempt All Three Questions.

[3 x 3 = 9]

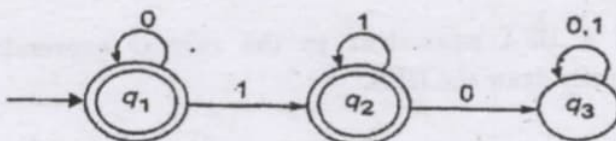
4. Construct a DFA for the set of strings over $\{a, b\}$ containing both **ab** and **ba** as substrings.
5. Construct a **Moore machine** that takes binary numbers as input and produces residue modulo '3' as output.
6. Let $\Sigma = \{a, b\}$. For each of the following languages over Σ , find a RE representing it.
 - a. All string that contain exactly one b
 - b. $L = \{w \mid w \text{ contains at least three consecutive 1s}\}$
 - c. All strings that contain either sub-string **aaa** or **bbb**.

Section - C

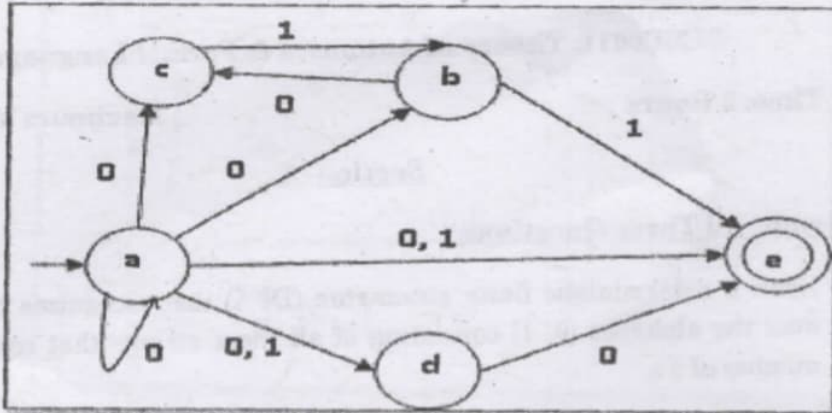
Note: Attempt Any Three Questions.

[3 x 5 = 15]

7. Apply Arden's theorem to find the Regular Expression corresponding to the following FA



8. Convert the following Non Deterministic Finite Automaton into an equivalent deterministic automaton M. Clearly mention all the 5 tuples of M and draw the complete transition graph.



9. Minimize the DFA whose transition table is given below. Draw the transition graph for the minimized DFA.

Present State	Next State	
	a	b
→q0	q1	q5
q1	q6	q2
q2 (Final state)	q0	q2
q3	q2	q6
q4	q7	q5
q5	q2	q6
q6	q6	q4
q7	q6	q2

10. Attempt both of the following.

[2.5 + 2.5 = 5]

- a. Consider a language over the alphabet {0, 1} consisting of the strings that meet the following conditions:
- The length of the strings is 6.
 - The last two characters must both be zero. For example, 110000, 001100, and 111100 are all in the language; 000011, 001010, and 111001 are not.

Write a regular expression that defines this language.

- b. Construct a DFA equivalent to the regular expression $a^* (ba^*)^*$.
Note: Directly draw the DFA.

Course Name: THEORY OF AUTOMATA & FORMAL LANGUAGES

Course Outcome

- CO1- Understand the basic concepts of Context Free languages, Expression and Grammars.
- CO2- Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.
- CO3- Analyze the process to convert regular expression to DFA, DFA to regular expression, and minimization of DFA.
- CO4- Develop the PDA for the context free language and context free grammar.
- CO5- Analyze that the grammar is ambiguous or unambiguous.
- CO6- Apply the process to convert CFG to CNF and GNF.
- CO7- Understand the concept of Turing machine and its variants.
- CO8- Design the Turing machine for the real world application.

Printed Pages: 02

University Roll No.

Mid Term Examination, Odd Semester 2022-23 B. Tech (CSE), III Year, V Semester Theory of Automata & Formal Languages (BCSC0011)

Maximum Marks: 30

Time: 2 Hours

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section - A

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Design a Finite Automata for $L = \{w \mid w \in \{a, b\}^* \text{ \& every } b \text{ is immediately followed by two } a\text{'s}\}$.	3	1	C	P
2	Find all strings in $L((a + b)^*b(a + ab)^*)$ of length three.	3	1	A	F
3	Give Regular Expressions for the following languages: a) L_1 = the set of all strings containing exactly 2a's. b) L_2 = {the set of all strings containing the substring bb, c) $L_3 = \{w \mid w \in \{0, 1\}^* \text{ and } w \text{ begins \& ends with the same symbol}\}$				
4	Construct a Moore machine that takes set of all strings over $\{a, b\}$ and counts no. of occurrences of substring 'baa'.	3	2	C	P
5	Construct a FA that accepts the strings having no. of 'a' divisible by 3 and no. of 'b' divisible by 2 i.e. $n_a(w) \equiv 0 \pmod{3}$, & $n_b(w) \equiv 0 \pmod{2}$, where $w \in (a, b)^*$	3	1	C	P

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E

Course Name: THEORY OF AUTOMATA & FORMAL LANGUAGES

Course Outcome

- CO1- Understand the basic concepts of Context Free languages, Expression and Grammars.
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Printed Pages: 02

University Roll No.

Mid Term Examination, Odd Semester 2022-23 B. Tech (CSE), III Year, V Semester Theory of Automata & Formal Languages (BCSC0011)

Time: 2 Hours

Maximum Marks: 30

Instruction for students:

1. All parts of a question should be answered at one place.
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Section – A

Attempt All Questions

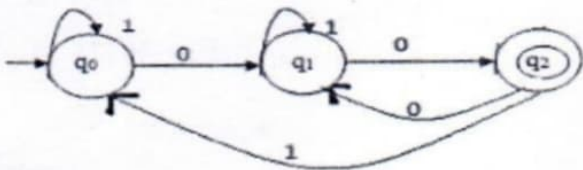
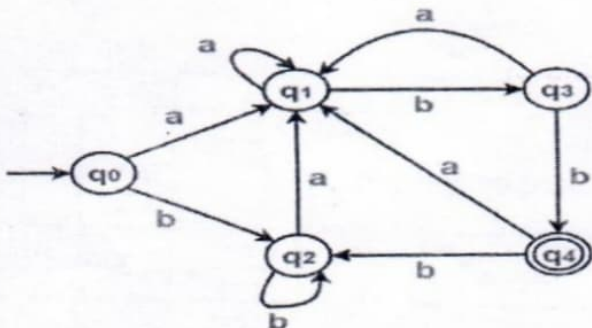
3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Design a Finite Automata for $L = \{w \mid w \in \{a, b\}^* \text{ \& every } b \text{ is immediately followed by two } a\text{'s}\}$.	3	1	C	P
2	Find all strings in $L((a + b)^*b(a + ab)^*)$ of length three.	3	1	A	F
3	Give Regular Expressions for the following languages: a) $L1$ = the set of all strings containing exactly 2a's. b) $L2$ = {the set of all strings containing the substring bb} c) $L3$ = $\{w \mid w \in \{0, 1\}^* \text{ and } w \text{ begins \& ends with the same symbol}\}$	3	3	E	C
4	Construct a Moore machine that takes set of all strings over $\{a,b\}$ and counts no. of occurrences of substring 'baa'.	3	2	C	P
5	Construct a FA that accepts the strings having no. of 'a' divisible by 3 and no. of 'b' divisible by 2 i.e. $n_a(w) \equiv 0 \pmod{3}$, & $n_b(w) \equiv 0 \pmod{2}$, where $w \in (a,b)^*$	3	1	C	P

Section - B

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
5	Design a PDA by using transition diagram for $L = \{ww^R \mid w \in (a,b)^+\}$.	5	4	C	M
7	State and prove Arden's theorem. Find the Regular Expression corresponding to the following FA using Arden's theorem. 	5	3	R & A	C & P
8	Minimize the DFA whose transition diagram is given below. Draw the transition graph for the minimized DFA. 	5	2	A	P

Mid Term Examination, Even Semester 2022-23**B.Tech (CSE), III Yr, VI Sem****BCSC0011: Theory of Automata & Formal Languages**

Time: 2 Hours

Maximum Marks: 30

Course Outcome

CO1: Understand the basic concepts of Context Free languages, Expression and Grammars.

CO2: Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.

CO3: Analyze the process to convert RE to DFA, DFA to RE, and minimization of DFA.

CO4: Develop the PDA for the context free language and context free grammar

CO5: Analyze that the grammar is ambiguous or unambiguous.

CO6: Apply the process to convert CFG to CNF and GNF.

CO7: Understand the concept of Turing machine and its variants.

CO8: Design the Turing machine for the real world application.

Instruction for students:

1. No overwriting should be there when designing machines.
2. Clearly mention if any assumption is made.

Section – A*Attempt All Questions*

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Attempt any one of the following. i. Draw the simplest possible DFA (in terms of number of states and arcs) that describes the language of all strings that end in "00". ii. Let $\Sigma = \{a, b\}$ and let $L = \{baa\}$. Design a DFA for L.	3	2	U	F
2	Use Arden's Theorem to find the Regular Expression that generates the same language as accepted by the following DFA. Clearly show all the steps.	3	3	A	P

3	Attempt <u>any one</u> of the following. - Write a regular expression that denotes the language of all strings over alphabet {a} whose length is not a multiple of 3. - Give a DFA for the following language: $L = \{w \in \{a, b\}^* \mid \text{the number of a's and the number of b's in } w \text{ have the same parity}\}$. Recall that two integers have the same parity if they are both odd or both even.	3	3	U	C
4	Attempt <u>any one</u> of the following. - Draw the NFA that recognizes the language where w contains the substring 0101. Do this using 5 states and assuming a binary alphabet. - Design a Mealy machine that scans sequence of 0s and 1s and generates output 'A' if the string read so far terminates in 00 and 'B' otherwise.	3	2	An	C
5	Give Regular Expressions for <u>any three</u> of the following languages - Set of strings of a's and b's ending with the string aabbaa. - Set of strings consisting of even number of a's followed by odd number of b's - All strings of 0's and 1's starting with 0 and ending with 1 - All strings of 0's and 1's with at least two consecutive 0's	3	3	U	F

Section – B

Attempt **All** Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Attempt any one of the following. i. Give the transitions for PDA for $L = \{a^n b^m c^n \mid m \geq 1, n \geq 1\}$. Also draw the transition graph. [3 + 2] ii. Design a PDA for $L = \{w \mid n_a(w) = n_b(w) \text{ i.e. the number of a's in } w \text{ is same as the number of b's}\}$. List all the transitions and also draw the transition graph. [3 + 2]	5	4	U	C
7	Convert the following NFA into equivalent Deterministic Finite Automata (DFA). Clearly show all the steps. There should not be any overwriting in the final table. <pre> graph LR start(()) --> q0((q0)) q0 -- a --> q0 q0 -- "a,b" --> q1((q1)) q1 -- b --> q1 q1 -- a --> q2(((q2))) q2 -- "a,b" --> q3(((q3))) q3 -- b --> q2 style start fill:none,stroke:none </pre>	5	2	A	P
8	Minimize the following DFA. Give the transition table and the transition graph of the minimized DFA.	5	3	A	M

Theory of Automata and Formal Languages (BCSC 0011)

Course Outcome

- CO1-Construct FA and minimize automata
- CO2-Construct an automaton for a given regular expression
- CO3-Construct a pushdown automaton for a given context-free language
- CO4-Construct a Turing machine deciding a given problem
- CO5-Prove whether a language is or is not a regular or context-free by using the Pumping Lemma
- CO6-Prove that a given context-free grammar generates a given context-free language
- CO7-Prove un-decidability of a given problem by reducing from a known un-decidable problem

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Printed Pages: 02

University Roll No.

Mid Term Examination, Odd Semester 2024-25

IIIrd Year, Vth Sem

Theory of Automata and Formal Languages (BCSC 0011)

Time: 2 Hours

Maximum Marks: 30

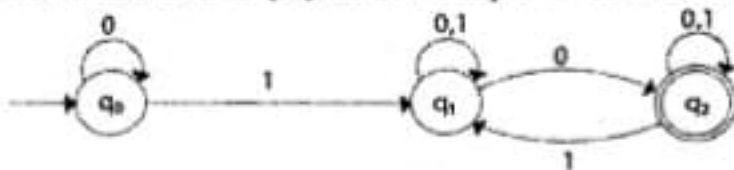
Instruction for students:

1. All parts of the question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section – A

Attempt All Questions

3 X 5 = 15 Marks

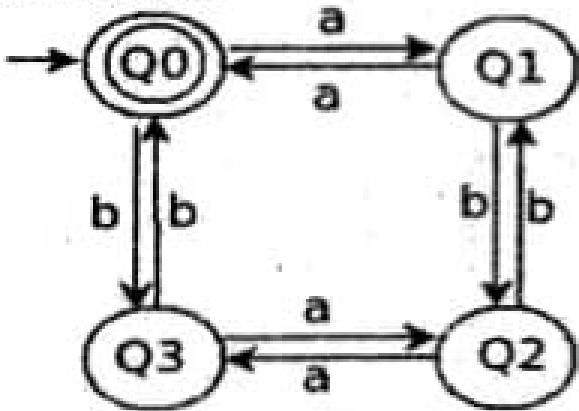
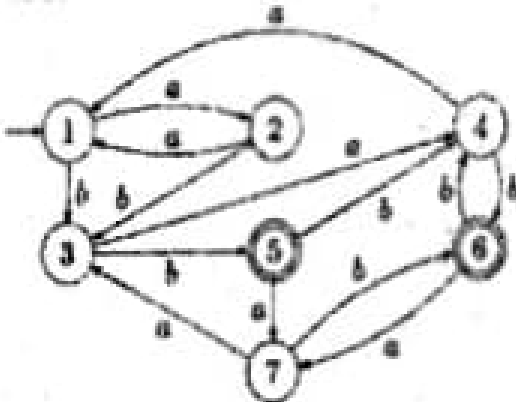
No.	Detail of Question	Marks	CO	BL	KL
1	Convert the NFA (N) to their equivalent DFA (M).  NFA (N)	3	CO1	R	P
2	Suppose we have to design a logic for a specific type of binary number in which the second last alphabet of the binary number is always "0", for example: xxxx0x, where x can be {0,1}. Create a DFA to implement the aforesaid problem.	3	CO5	U	F
3	Create a DFA with the help of null removal for the regular expression (R1). R1 = (a+b)*(aa+bb)(a+b)*	3	CO2	A	C
4	Design a Moore machine which determines the residue modulo 3 for each binary string treated as binary integer.	3	CO1	An	M
5	Write down the Regular expression for the following: (a) The language starting with a but not having consecutive b's. (b) $a^m b^{2n} c^{3p} \mid m, n > 0 \text{ and } p \geq 0$.	3	CO2	U	F

(c) The set of strings over $\{0,1\}$ that have at most one 1.

Section – B

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Find out the regular expression of the given DFA (M) using Arden's theorem:  DFA (M)	5	CO2	A	P
7	Design a PDA which accept the string of the language (L), where $L = \{ w c w^R \mid w = \{a, b\}^* \}$	5	CO3	C	C
8	You are a software engineer working on an email filtering system for a large organization. The system uses a DFA (M) to classify incoming emails based on specific patterns to determine if they should be flagged as spam. The goal is to simplify the DFA (M) using Equivalence Theorem used in the filtering system to enhance its performance and reduce processing time.  DFA (M)	5	CO1	An	P

Course Name:

Course Outcome

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Printed Pages: 03

University Roll No.

Mid Term Examination, Even Semester 2024-25

B.Tech (CSE/AIML/CCV/CSF/DA/IIoT), III Year, VI Semester

BCSC0011: Theory of Automata & Formal Languages

Time: 2 Hours

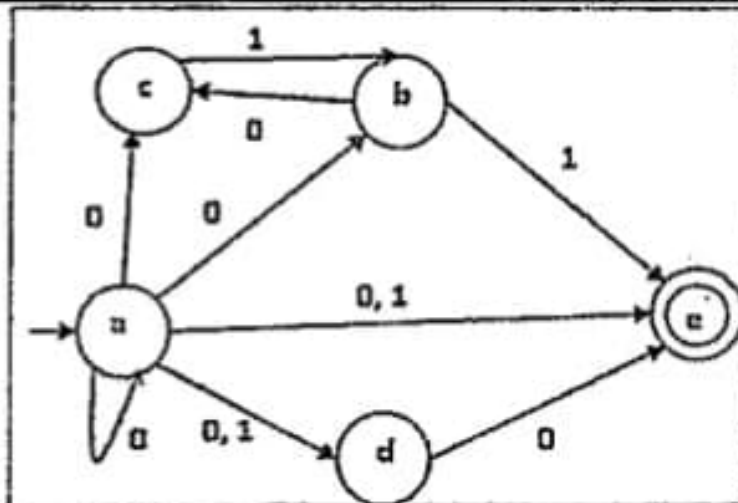
Maximum Marks: 30

Section – A

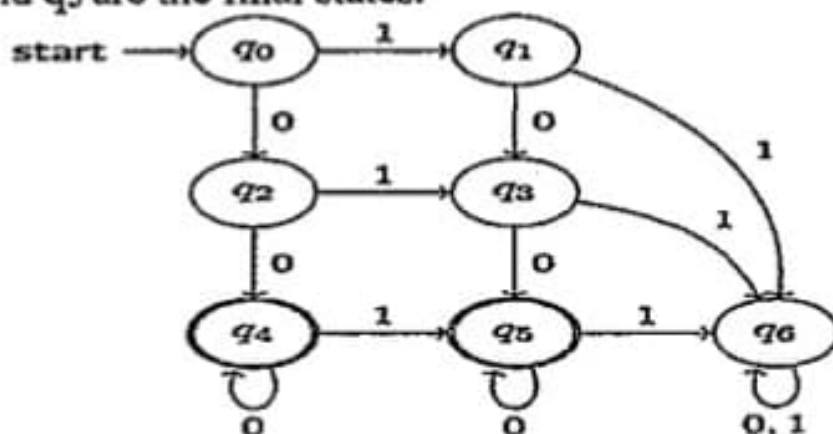
Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	K L
1	Write the Regular expressions for the following languages: a. A string that contains exactly two zeroes over input alphabet $\{0, 1\}$. b. A string of length at least 1 that starts and ends with the same character, where the alphabet $\Sigma = \{a, b, c\}$	3	CO3	A	C
2	Construct a Mealy Machine that outputs a binary string that contains a 1 for every 'aa' substring in an input string composed of a's and b's. For example, if 'aaabab' is the input string, 011000 is the corresponding output.	3	CO2	An	C
3	Convert the given Nondeterministic Finite Automaton (NFA) into its equivalent Deterministic Finite Automaton (DFA).	3	CO2	A	P



Define a Deterministic Finite Automaton (DFA) and show the transition steps to check if the following DFA accepts the string 0100010. Here q_4 and q_5 are the final states.

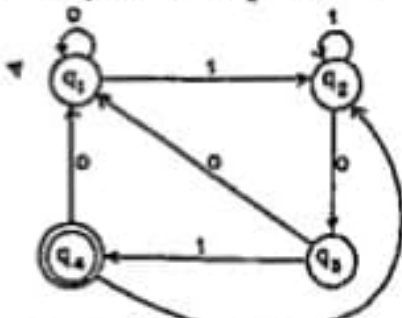
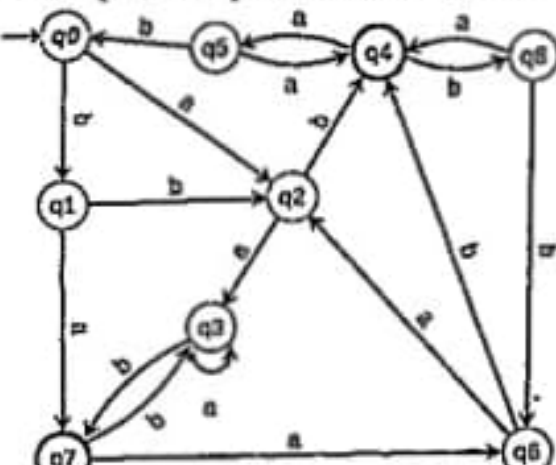


Suppose you are using an online game platform that accepts the power characters a and b as in-game power to unlock a new ability. The ability is unlocked if the power character occurs in the sequence accepted by the following regular expression.

$$((aa + b)(a + b) + ab)(a + b)^*$$

For example,

If power character sequence is aaa, or ab, or bb, or aaaa, etc., the ability is unlocked. Draw the DFA that accepts the above regular expression using null removal

No	Detail of Question	Marks	CO	BL	KL
6	State & prove the Arden's theorem. Find the regular expression corresponding to the following FA using Arden's theorem (assume q_1 as initial state): 	5	CO3	R	C
7	Minimize the following DFA using K-equivalence method where q_4 and q_7 are final states: 	5	CO3	U	P
8	Enlist the tuples of PDA with their meanings. Suppose a packing machine is designed to wrap items in two colors: red and green. The machine receives a sequence of colors followed by the wrap command as input and starts wrapping the items and outputs the wrapped item in reverse order of the given sequence of colors. For example, if the input is red, green the machine will wrap the items in the order green, red after receiving the wrap command. Design a Pushdown Automaton (PDA) that should check if the items obtained at the output have been wrapped in the correct reverse order or not. Also, show the transition steps to accept the following packing order: red, green, wrap, green, red	5	CO4	An	M